REBOOT THE EARTH 2026 / GROUP 6

Verdifica

Making Forest Product Traceability More Accessible

A digital green passport concept supported by field records and satellite evidence

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TECHNICAL PROJECT REPORT

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github.com/victoraugustotlb/Reboot-the-Earth-Grupo6

Developed during Reboot the Earth 2026

We built Verdifica as an open-source hackathon prototype for small-scale forest producers in the Brazilian Amazon, who may face barriers when documenting timber origin, organizing extraction records, and accessing formal supply chains. Our goal is to make the preparation and sharing of a digital green passport more accessible through product records, traceability identifiers and satellite-based environmental screening. It integrates a React interface, a Node.js/Express application and SQLite database, and a Python/FastAPI geospatial service using Google Earth Engine. This report documents the producer journey, the software we built, the geospatial methods we tested, and a January–October 2026 demonstration-area export of Sentinel-2 imagery, NDVI summaries, and Dynamic World land-cover visualizations. Mean monthly NDVI ranged from 0.552 (August) to 0.709 (January); the data do not establish deforestation, product origin, or legal compliance. The token and QR workflow is experimental; official permit verification, independently validated product provenance, and public end-to-end traceability remain outside the scope of our prototype.

Keywords: forest traceability; digital public infrastructure; remote sensing; Sentinel-2; NDVI; Dynamic World; environmental evidence.

The Challenge: Reboot the Earth

We developed Verdifica during the *Reboot the Earth* hackathon, held with United Nations participation. We worked on the challenge **"AI-Powered Digital Public Infrastructure for Sustainable and Traceable Forest Supply Chains."** The challenge emphasized three complementary capabilities: (1) AI-supported geospatial intelligence using satellite and land-cover information; (2) digital traceability and product passports connecting supply-chain actors; and (3) decision support combining geospatial signals, permits and supply-chain records.

Our response was to prototype a workflow in which a small producer can declare a product and location, associate photographic evidence, request an environmental screening, and receive a reviewable report. We chose to focus on the gap between a producer's field records and the documentation a buyer or inspector needs to examine. Our prototype addresses that gap without claiming to replace official authorization or inspection.

Why We Built Verdifica

Who we designed for

We designed Verdifica primarily for the **small-scale forest producer in the Brazilian Amazon**: someone whose livelihood may depend on responsible forest management or timber extraction but who does not necessarily have access to specialized GIS software, compliance teams, or costly digital traceability systems. We chose timber as our first demonstration case because our challenge centered on forest-product origin and traceability. We envision adapting the approach to other forest products, but we have not validated that extension.

The pain point: evidence is difficult to assemble and communicate

Even when an activity is authorized, a producer may struggle to organize the information needed to demonstrate where a product came from, associate it with the relevant permit, and share trustworthy records with buyers or reviewers. Fragmented paperwork and limited digital access can become barriers to formal markets. Our presentation identified four connected problems:

vulnerability, bureaucracy, exclusion from formal markets, and the undervaluation of responsible forest management. We designed for these constraints; we have not yet measured their prevalence or the economic effect of our solution.

The goal is **not to let an algorithm declare a shipment legal**. It is to reduce the friction of assembling records and to provide understandable environmental context for further verification. Public authorities remain responsible for official legal determinations.

A concrete example

We pictured a producer preparing a timber batch for sale. Rather than keeping a photograph, location pin and authorization in separate places, they could create a digital record linking product information, an evidence capture, a declared geographic area and a reference to the applicable authorization. Verdifica can then request a satellite-based screening and organize its output into a report. A buyer or inspector could then see the declared origin and the available evidence, alongside a clear account of what our screening did and did not verify.

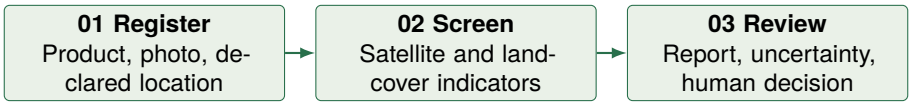


Figure 1: Producer-centered journey, simplified from the team's original Reboot the Earth presentation. It represents the intended workflow; real-world legal validation is not automated.

Making the Green Passport Practical

For us, the starting point was simple: a small producer should not need a dedicated compliance department or expensive mapping software just to organize the evidence behind a timber batch. A photograph alone does not explain where wood was collected; a location alone does not establish authorization; and a permit stored separately from the batch record is difficult for a buyer to interpret. We designed Verdifica to bring those pieces together in a single, readable trail.

We use the term *green passport* to describe a **digital traceability record**, not a government-issued certificate. Our proposed passport connects the batch identifier, product description, declared origin, photographic evidence, references to relevant authorizations, geospatial screening results and the date and status of the analysis. We want this process to be **simpler, less costly and less bureaucratic**. We have not conducted producer trials or measured savings, so these are design goals rather than proven outcomes.

From a forest product to a shareable record

We organized the intended journey around five actions:

1. **Register a batch.** The producer enters the product details and records a photograph and declared location in the application.
2. **Associate evidence.** The record keeps the capture attempt, geographic reference and any authorization information available for review.
3. **Request screening.** Our FastAPI service processes satellite-derived vegetation and land-cover indicators and compares them with a declared permission geometry when provided.

4. **Read the report.** The application organizes the screening status, maps and numerical indicators into a report. A flagged inconsistency calls for human review; it is not a finding of illegality.
5. **Share a traceability reference.** An experimental token and QR-code flow can connect the record to its evidence, providing the basis for a future consultation experience for buyers, consumers and inspectors.

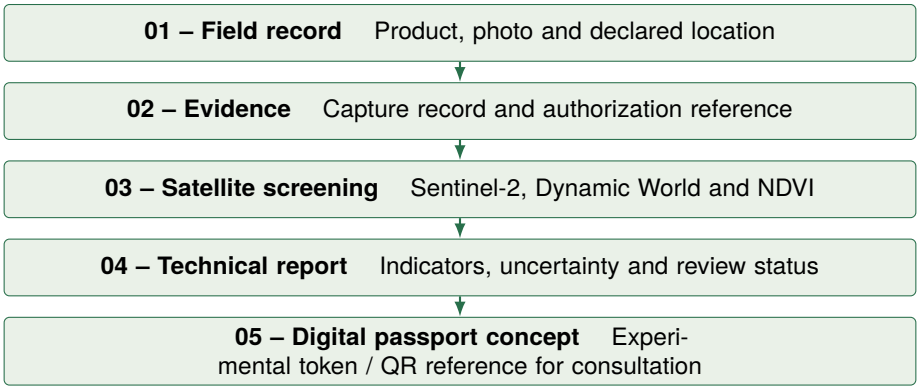


Figure 2: Our proposed green-passport journey. The hackathon prototype implements the main record and screening components; a verified public passport and complete chain of custody require further development.

Who would consult the passport?

We designed the passport for three different moments: the **producer** assembles and shares the batch record; a **buyer or consumer** follows the identifier to the declared provenance and supporting evidence; and an **inspector** reviews the technical report against official records. The first steps are represented in our prototype. A verified, publicly accessible consultation service across the supply chain is still future work.

What we are trying to remove is administrative friction, not legal safeguards. Verdifica does not replace extraction permits, chain-of-custody audits or the authority of environmental agencies. It aims to make the supporting information easier to collect, organize and check.

The Product: What Verdifica Does

We built the application to organize product records, capture attempts, locations and photographs. Our prototype combines a React frontend with a Node.js/Express API and SQLite persistence. A separate Python/FastAPI service processes geospatial indicators and creates technical reports. Records and evidence are linked to an experimental traceability workflow, including QR labels and demonstration tokens.

Important boundaries. The demonstration token is not a legal approval or an independently verified provenance credential. The photograph-checking endpoint validates file characteristics but does not establish image authenticity, embedded GPS provenance, or a trained computer-vision classification. The local MVP does not provide automatic offline synchronization; this remains a design goal.

What we built, and what remains to be tested

Table 1: Scope of the hackathon prototype.

Working prototype	Next validation step
Product records, photos and location capture	Test the workflow with producers in field conditions.
Satellite screening and technical reports	Validate indicators against independent reference data.
Demo token and QR labels	Build a secure, verified public passport lookup with access controls.
Demo permits and review statuses	Integrate authoritative permits and human regulatory review.

How We Built It

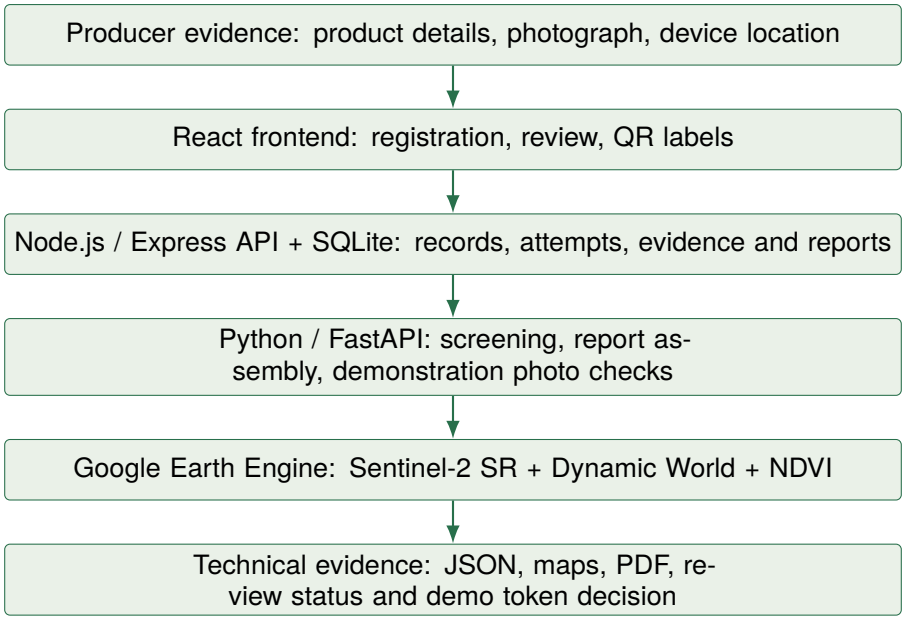


Figure 3: Our prototype architecture. The diagram describes implemented components, not an audited production deployment.

Geospatial decision support

The geospatial module evaluates vegetation and land-cover indicators, including candidate changes inside and outside a declared permission geometry when one is supplied. Permission records in the demonstration are simulated. The application applies conservative rules to demonstration token eligibility: missing or invalid evidence, stale analysis, errors, and adverse or uncertain results do not constitute approval.

Engineering considerations

The local application has integration tests and a documented optional API-token configuration. It is not a security-audited production system; authentication is disabled by default in the local MVP. Product photos, GPS coordinates and producer information require careful access control before public deployment. We also need to protect sensitive location and producer data: a public

consultation view should expose only the minimum information necessary, and QR payloads must not reveal private details.

How the AI and Geospatial Screening Work

Machine learning in the actual implementation

We **integrated an existing machine-learning-based geospatial product** rather than training a new forest-classification network. *Dynamic World* provides near-real-time land-cover probabilities and labels derived from Sentinel-2 imagery. The geospatial service combines these outputs with Sentinel-2 surface reflectance, NDVI calculations, time comparisons and declared area geometries. A rule-based screening layer turns the spatial indicators into a status requiring review when potential inconsistencies are detected.

This distinction matters: **Dynamic World classification is ML-derived**; NDVI is a spectral index; the threshold and comparison logic are programmed decision rules. None of these alone verifies a harvest permit, the physical origin of timber or the authenticity of a producer's photograph.

From pixels to a report

The workflow begins with the coordinates or polygon supplied for a product record. The geospatial service requests Earth Engine observations for the area and time window, masks unsuitable pixels where implemented, calculates vegetation indices and inspects land-cover changes. If a demonstration authorization geometry is provided, candidate changes may be separated into areas inside and outside that polygon. The service then assembles numerical indicators, map references, caveats and a review-oriented status into JSON and PDF report pathways.

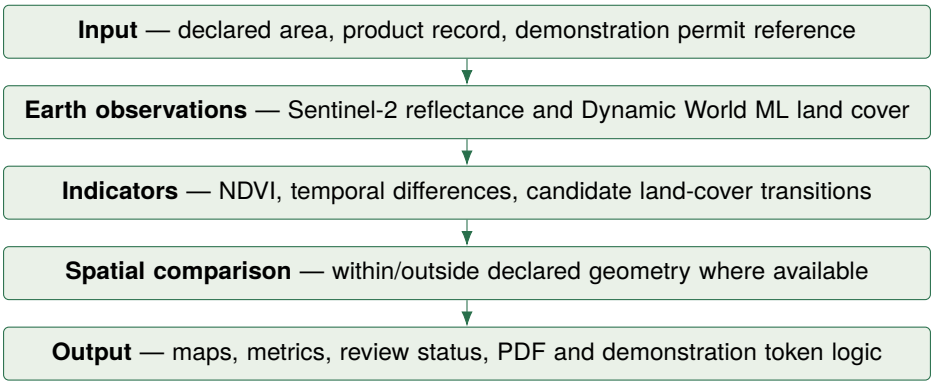


Figure 4: Geospatial processing stages implemented in our FastAPI service. Some components use fictional demonstration authorizations.

Why the Amazônia-1 experiment matters

Alongside Sentinel-2, we tested Brazil's Amazônia-1 WFI satellite to explore national Earth-observation data for forest monitoring. Earlier experiments produced an NDVI time series and before/after comparisons for 2024–2025. They demonstrate an alternative source of observations but use different processing assumptions, spatial resolution and cloud-handling limitations. For that reason, we use the 2026 Sentinel-2 export for the main figures and keep Amazônia-1 as a separate experiment.

Photo analysis: what was and was not implemented

We included a demonstration photo-analysis endpoint that checks basic file properties. It does *not* perform operational visual authenticity verification, forensic tamper detection or reliable GPS extraction from the image itself. The app's captured device location is distinct from GPS metadata embedded in a photo. These checks would require a future implementation and validation before being described as AI-based provenance verification.

Satellite Monitoring: Data and Method

Demonstration area. The 2026 export is centered at latitude -12.5, longitude -54.5 (Mato Grosso, Brazil), using a nominal 1 km analysis radius. This is a test area, not proof of a registered producer's property or a field-verified forest-product origin.

Sources. Monthly images were exported from the Sentinel-2 surface-reflectance collection and the Dynamic World land-cover collection in Google Earth Engine. The dataset covers January–October 2026. The exported RGB images are visual median composites without per-pixel SCL cloud masking; NDVI processing uses Sentinel-2 scene-classification (SCL) masking. The Dynamic World images are model-derived land-cover visualizations, not independently verified ground-truth maps. No custom machine-learning model was trained for this export.

Vegetation indicator. Normalized Difference Vegetation Index is defined as

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{red}}}. \tag{1}$$

Monthly mean NDVI was calculated for available valid pixels. Cloud conditions, vegetation phenology, sampling differences and composite construction affect interpretation. A change in mean NDVI is not equivalent to deforestation or illegal extraction.

2026 Visual Atlas

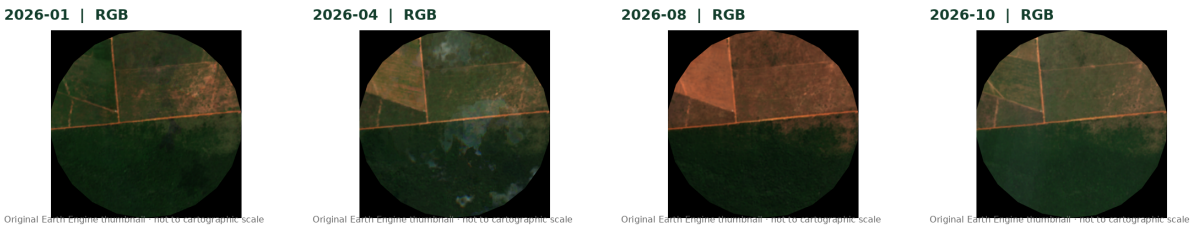


Figure 5: Sentinel-2 RGB thumbnail composites for January, April, August and October 2026. These visualizations come from our 2026 export; they are *not* georeferenced map products and therefore carry no invented scale bar.

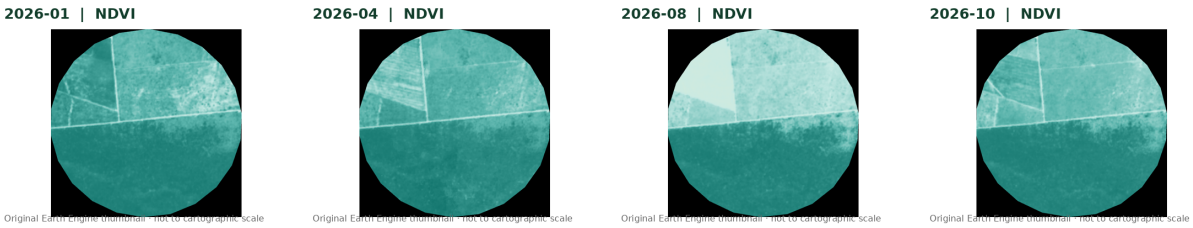


Figure 6: NDVI visualizations for the same four months. Colors follow the original export; without the export's exact visualization parameters, no quantitative colorbar is inferred from thumbnail colors.

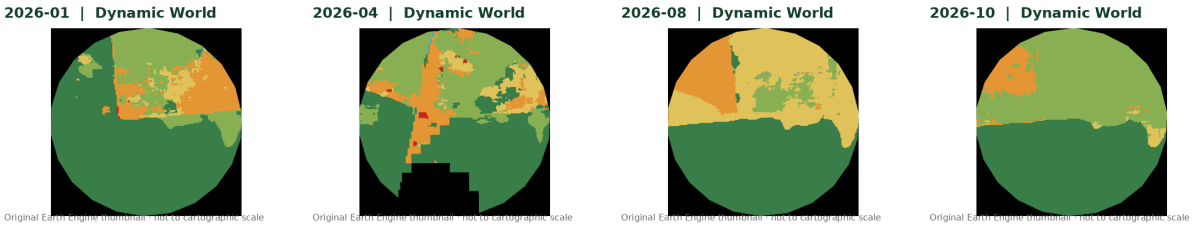


Figure 7: Dynamic World land-cover thumbnails for four months in 2026. Class colors are visualization outputs, not an accuracy-validated local land-cover survey.

Cartographic note. Our exported PNG files do not retain a coordinate reference system or pixel-to-ground transform. A precise north arrow, coordinate grid or kilometer scale bar cannot be reconstructed reliably from them. To add cartographic scale in a later revision, export GeoTIFFs with CRS, transform and pixel resolution (or retain original Earth Engine export geometry).

Measured Indicators

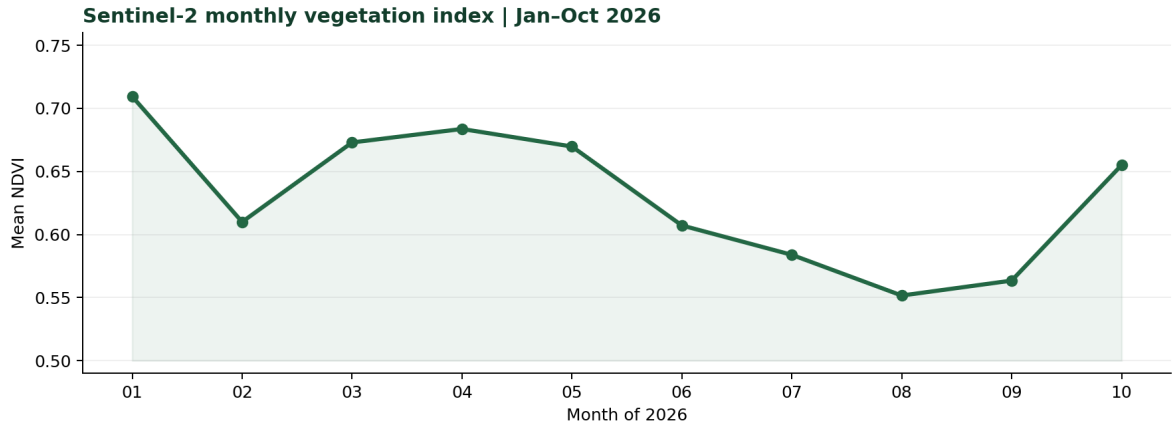


Figure 8: Mean monthly NDVI from our `monthly_metrics.csv`. January has the highest reported monthly mean (0.709), while August has the lowest (0.552).

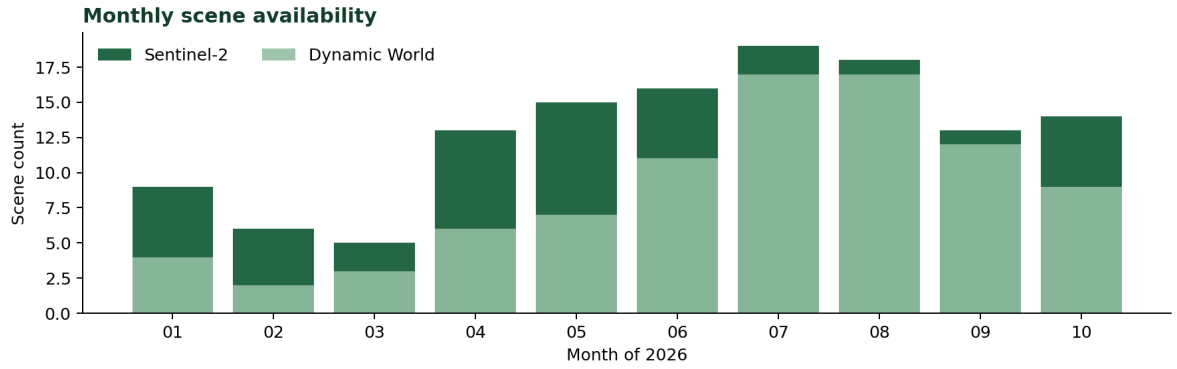


Figure 9: Monthly counts of Sentinel-2 and Dynamic World scenes included in the export metadata. These are scene counts, not the number of independently validated cloud-free observations.

Table 2: Selected values from the January–October 2026 export.

Month	Mean NDVI	Sentinel-2 scenes	Valid NDVI area (ha)
January	0.709	9	309.2
April	0.684	13	309.2
August	0.552	18	309.2
October	0.655	14	309.2

Across all ten months, the export reports **128 Sentinel-2 scenes**. The mean NDVI declines between January and August, then increases by October. This may reflect seasonal vegetation behavior, observation conditions, or other changes; it is *not* evidence on its own of clearing or restoration. February reports a smaller valid NDVI area (266.0 ha) than the other months (309.2 ha), highlighting that the usable spatial sample is not identical throughout the series.

What the Prototype Demonstrated

We assembled a multi-component application with product and evidence registration, persistence, a geospatial screening service, and report-generation pathways. Our 2026 export demonstrates the practical ability to retrieve monthly Sentinel-2 and Dynamic World imagery and summarize NDVI over a chosen area. The engineering result was the integration of existing technologies into a working demonstration, not the training of a new satellite model.

Earlier Amazônia-1 work. We also explored Brazil's Amazônia-1 WFI sensor in separate 2024–2025 experiments. Those historical results used different data and processing assumptions and are deliberately not combined numerically with the 2026 Sentinel-2 series. They remain part of the project history, not the primary evidence for this report.

How the Prototype Responds to the Three Challenge Tracks

Table 3: Relationship between the hackathon challenge and the demonstrated prototype.

Challenge track	Verdifica response
A. AI-supported origin and forest risk	Earth Engine screening using Sentinel-2, NDVI and ML-derived Dynamic World land cover. Preliminary signals, not verified product origin.
B. Product traceability and digital passports	Producer records, photos, identifiers, QR/demo token concepts and persistent application data. End-to-end multi-actor chain of custody remains future work.
C. Compliance and risk decision support	Demonstration permit comparison, review statuses and PDF evidence pathways. Official permit integration and legal certification are not implemented.

Why This Matters: Our Contribution to the SDGs

We designed Verdifica around a social and environmental tension: small producers need a realistic path into transparent formal markets, while buyers and public authorities need evidence they can inspect. We see a potential connection to four United Nations Sustainable Development Goals (SDGs), which we highlighted in our hackathon presentation.

Table 4: How our design relates to the Sustainable Development Goals. These are intended contributions, not measured outcomes.

Goal	Connection to Verdifica
SDG 8 – Decent Work and Economic Growth	We want smaller forest producers to have a more accessible way to present organized records to formal buyers, potentially reducing barriers to market participation.
SDG 12 – Responsible Consumption and Production	We propose a digital passport so product information and supporting evidence can follow a batch and become easier for buyers and consumers to inspect.
SDG 13 – Climate Action	We use Earth-observation indicators to support earlier examination of potential vegetation changes; we have not measured avoided emissions.
SDG 15 – Life on Land	We connect traceability with forest-cover screening to support responsible forest management and human review of possible inconsistencies.

Our immediate contribution is a prototype that connects field records with environmental context. The broader promise—lower documentation costs, more transparent supply chains and better-informed checks—still needs testing with producers, buyers and environmental authorities. We do not claim that the prototype has already reduced illegal extraction, administrative expenses or forest loss.

The Team and the Hackathon Experience

We developed Verdifica as Group 6 at Reboot the Earth 2026: **Giovanna Franco de Carvalho, Victor Augusto Toledo Lúcio Borghi, Thales Prado Rocha dos Santos, Rafael Hoki Kuntz and Thiago Silva de Almeida**. Together, we worked across product design, interface development, application engineering and geospatial analysis.

We worked within a hackathon timebox, prioritizing a demonstrable end-to-end prototype over a production-ready compliance system. In our final demonstration, we presented the producer

registration interface, a satellite-supported reporting concept and an experimental token decision. The most important lesson for us was to keep the producer's experience at the center while being precise about what satellite data can support and what still requires human verification.

Limitations, Responsibility and Next Steps

The analysis lacks independent field ground truth, model accuracy measurement and a fully georeferenced 2026 raster deliverable. A demonstration permission is not an official authorization. Satellite signals cannot confirm a product's origin, determine legal compliance or identify the cause of spectral change. The system should be interpreted as a preliminary screening and evidence-organization prototype for human review.

A next iteration should export georeferenced rasters, preserve processing parameters, evaluate cloud and shadow handling, validate change detection against reference samples, connect to authoritative permits, strengthen access control and QR-label privacy, and test producer workflows in real conditions. In our hackathon presentation, we proposed species identification with ML, extension to other Amazonian forestry products and expansion to neighboring Amazon countries. Those items are future research and product directions, not demonstrated features. Offline-first synchronization and interoperable governance remain important longer-term objectives.

Closing Perspective

We built Verdifica around a practical question: how can accessible software help smaller producers organize traceability evidence while giving reviewers better environmental context? Our prototype offers one modular answer, combining familiar web technologies with open Earth-observation datasets and pre-trained geospatial products. Our next task is to test whether this workflow genuinely saves producers time, helps reviewers assess evidence and can be operated responsibly beyond a hackathon.

Project Links and Data Provenance

Repository: `https://github.com/victoraugustotlb/Reboot-the-Earth-Grupo6`
Data: January–October 2026 demonstration export (Sentinel-2, NDVI and Dynamic World); underlying values are included in `monthly_metrics.csv`.
Implementation: Project source code and README in the repository linked above.
Project presentation: *Verdifica: Extrativismo Sustentável & Controle de Qualidade*, presented during Reboot the Earth 2026.
Further reading: Sentinel-2 dataset; Dynamic World dataset; Google Earth Engine documentation.